

## NMR Transmit–Receive System with Short Recovery Time and Effective Isolation

K. JURGA,\* E. C. REYNHARDT, AND S. JURGA\*

*Department of Physics, University of South Africa, P.O. Box 392, Pretoria 0001, South Africa*

Received May 2, 1991

A transmit–receive system with a short recovery time and excellent isolation has been developed. The system operates in conjunction with an ENI Model 3200L broadband amplifier and a spin-lock NMR pulse spectrometer. The system has been tested in the frequency range 5.5 to 52 MHz and seems not to generate any background noise. © 1992

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A well-designed transmit–receive system should meet several requirements. (a) During the transmitting mode, energy should be transmitted to the sample coil while the receiver should be effectively isolated from the strong RF signal. Isolation plays a crucial role, especially in  $T_{1\rho}$  experiments, where the gain of the receiver must be constant and independent of the length of the spin-locking RF pulse. The amplitude of the spin-locking pulse should be constant for a period of up to 250 ms. (b) With a view to detecting the weak NMR signal with a maximum signal-to-noise ratio, the tuning box and sample coil should be isolated during the receiving mode from noise originating from the transmitter. In addition, the recovery time should be as short as possible.

The system described in this paper has been designed to eliminate a number of problems which were encountered during the modification of a spin-lock pulse spectrometer operating at 5.5, 20, 30, and 52 MHz. This spectrometer is an excellent instrument, but its weak point is the final stage of the transmitter. It was decided to replace this final stage with an ENI Model 3200L broadband transmitter. The transmit–receive system described in this paper fulfills both the above-mentioned prerequisites.

### GENERAL DESCRIPTION OF THE SYSTEM

Since a single pair of crossed diodes, aimed at isolating the ENI transmitter from the tuning box, was not sufficient to eliminate noise and the leakage of a low-frequency spurious signal from the transmitter to the receiver input, an isolation circuit, hereafter referred to as the RF high-voltage pass (HVP), was designed. As shown in Fig. 1, this isolation circuit is introduced between the output of the ENI amplifier and the input of the tuning box and sample coil (TBSC). This modification allows the use of the ENI power amplifier in the pulse mode.

It was found that a  $\lambda/4$  cable terminated by crossed diodes (1) did not lead to a constant receiver gain as a function of the spin-locking pulse length. The problem of

\* Permanent address: Institute of Physics, A. Mickiewicz University, 60-780 Poznan, Poland.

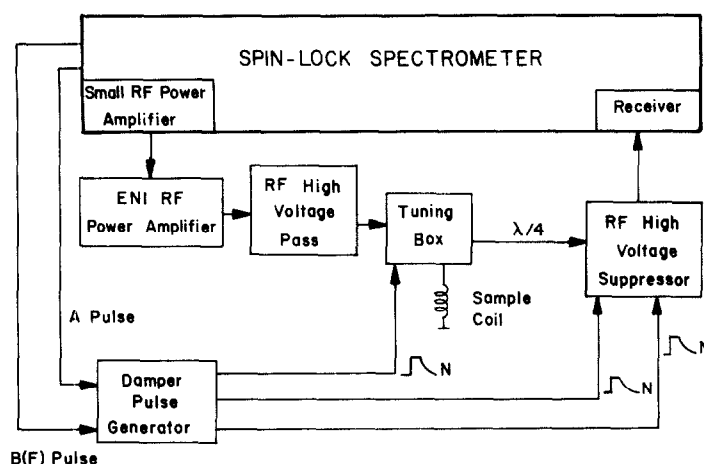


FIG. 1. Diagram of the transmit-receive system with damper pulse generator for the reduction of recovery time.

isolation has been addressed at some length in the literature (2-5), but for the frequency range of 5.5 to 52 MHz and transmitter power of about 200 W, the available designs are all unsuitable. The disadvantage of the coupling scheme proposed by Kisman and Armstrong (2) is that the diodes conduct when the weak signal is observed, thus introducing some noise. Hoult and Richards (3) developed an interesting receiver protection scheme which is convenient in the case of ultrahigh frequencies. The lumped circuit duplexer proposed by McLachlan (4) proved to be inadequate in our case. Engle (5) designed a low-noise broadband receiver circuit, but it is limited to low power ( $<100$  W).

With a view to overcoming this problem, an RF high-voltage suppressor (HVS), which functions in conjunction with a  $\lambda/4$  cable and crossed diodes, was designed. The HVS gives strong attenuation of the strong signal, but the attenuation of the weak NMR signal is negligible.

The recovery time of the spectrometer was reduced by employing a previously described method (6). In addition, a modified damper pulse generator was used. When the HVS is driven by the damping pulse, the troublesome weak signal at the end of the RF pulse is reduced.

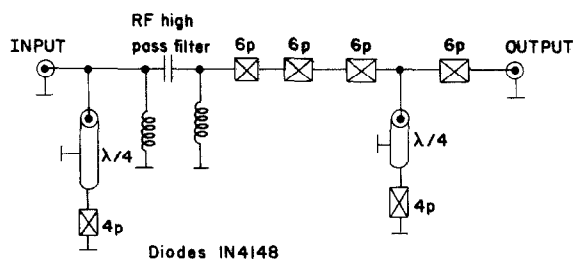


FIG. 2. Circuit diagram of the RF high-voltage pass.

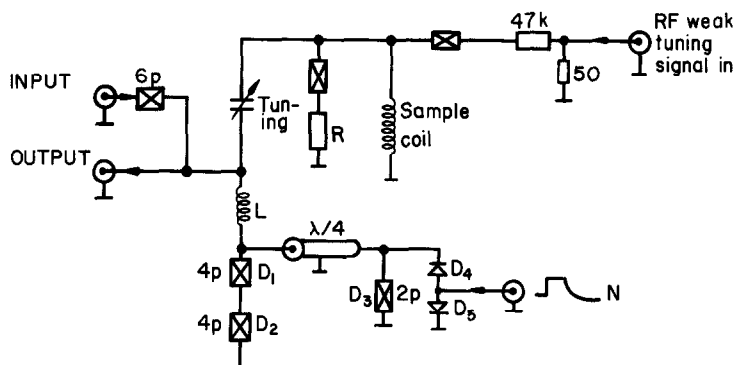


FIG. 3. Circuit diagram of the tuning box, sample coil, and  $\lambda/4$  cable for the reduction of the recovery time.

A block diagram of the transmit-receive system, together with the damper pulse generator, is shown in Fig. 1. A small RF signal from the output of the spin-lock spectrometer is amplified by the ENI power amplifier and the resulting strong RF signal is transmitted through the HVP to the TBSC. The output of the latter is connected by means of a  $\lambda/4$  cable to the HVS. The output of the HVS is connected to the input of the spin-lock receiver. With a view to reducing the recovery time, pulses N from the damper pulse generator are transmitted to the TBSC and HVS.

#### CIRCUIT DIAGRAMS

*RF high-voltage pass.* The circuit diagram of the HVP is shown in Fig. 2. The essential component of the system is a high-pass filter which suppresses the spurious

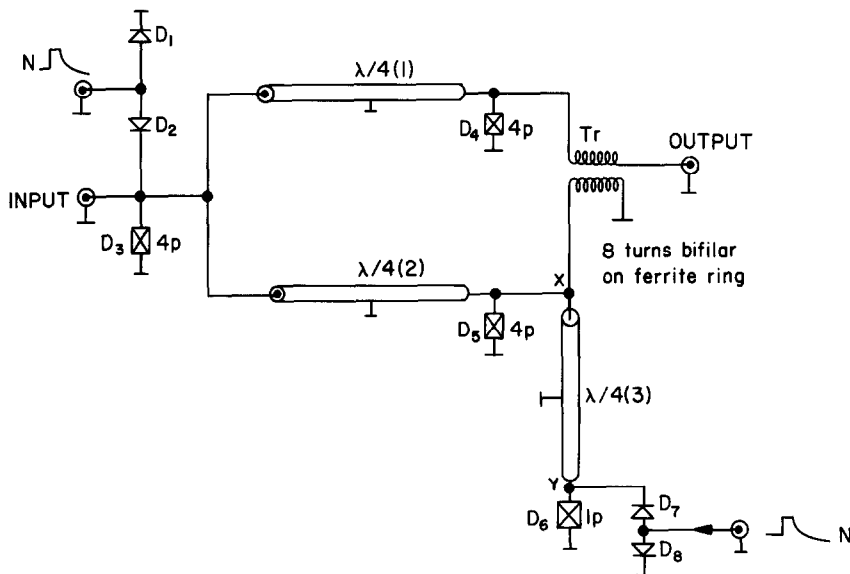


FIG. 4. Circuit diagram of the RF high-voltage suppressor.

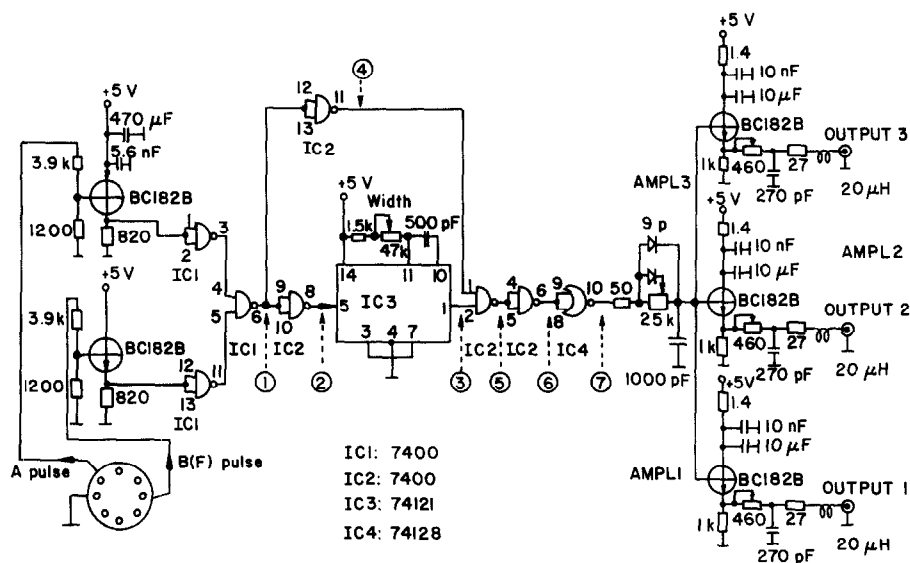


FIG. 5. Logic diagram of the damper pulse generator.

low-frequency signal from the ENI transmitter. Two  $\lambda/4$  cables terminated by crossed diodes attenuate the weak RF signal (7, 8). At 52 MHz the attenuations of the weak and strong signals are 51 and 0.4 dB, respectively.

*Tuning box and sample coil.* The circuit diagram for the TBSC is shown in Fig. 3. The coil L is adjusted to obtain the 50  $\Omega$  output impedance for a weak signal. Resistor R damps the high-voltage part of the ringdown and the  $\lambda/4$  cable, together with crossed diodes  $D_1$ ,  $D_2$ ,  $D_3$ ,  $D_4$ , and  $D_5$ , reduces the recovery time. This represents a modification to the circuit described previously (6).

*RF high-voltage suppressor.* Figure 4 shows the circuit diagram for the RF high-voltage suppressor. This circuit attenuates a strong signal very effectively, but transmits a weak signal with very low attenuation if the N damping pulses are switched off. If these pulses are on, the weak signal is attenuated, leading to a reduction of the recovery time. Let us consider the behavior of the circuit for strong and weak signals in the absence of the N pulses. When a strong signal, usually a few volts, is fed to the input of the circuit, the diodes  $D_3$ ,  $D_4$ ,  $D_5$ , and  $D_6$  conduct and therefore the voltages at the outputs of the  $\lambda/4(1)$  and  $\lambda/4(2)$  cables are the same. Consequently these voltages are cancelled by the transformer Tr. However, if a weak signal is fed to the input of the circuit, with the N pulses off, the diodes  $D_3$ ,  $D_4$ ,  $D_5$ , and  $D_6$  do not conduct. The impedance at point X becomes low (X is grounded), while the impedance at the input of the  $\lambda/4(2)$  cable becomes large. The weak input signal is therefore transmitted via cable  $\lambda/4(1)$  and reaches the output of the circuit almost unattenuated. Next the influence of the N pulses on the operation of the circuit will be considered. As mentioned above, the weak signal is attenuated. The point Y is grounded and the  $\lambda/4(3)$  cable has a large impedance at X. The weak signal is therefore also cancelled by the transformer Tr. It is emphasised that, during the duration of the weak NMR signal, all the diodes are nonconducting and therefore no noise is introduced to the system.

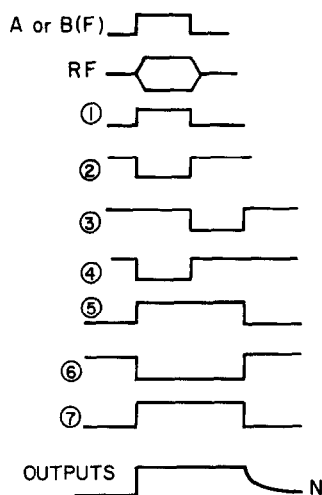


FIG. 6. Timing of the damper pulse generator.

*Recovery time.* As mentioned above the recovery time was reduced by employing a modification of a method published previously (6). Instead of using one N pulse, three N pulses are used. The amplitude of each of these pulses can be regulated individually, while the widths and slopes can be regulated simultaneously. The damper pulse generator is shown in Fig. 5. The timing of the pulses generated by the damper is shown in Fig. 6.

#### CHARACTERISTICS

Attenuation of a strong signal at 52, 30, and 5.5 MHz is 25.7, 26.8, and 11.9 dB, respectively. Attenuation of a weak signal with the N pulses off is 1.3, 0.07, 1.4, and 2.7 dB at 52, 30, 20, and 5.5 MHz, respectively. The introduction of the HVS makes it possible to measure relaxation times in the rotating frame at constant gain of the receiver for  $100 \mu\text{s} < \tau < 250 \text{ ms}$ , where  $\tau$  is the length of the spin-locking pulse.

Proper regulation of amplitudes, widths, and slopes yielded a recovery time of 12  $\mu\text{s}$  at 5.5 MHz with a  $Q$  value of 60 for the loaded circuit. For the same  $Q$  value, recovery times at 20 and 30 MHz were found to be 8 and 7.5  $\mu\text{s}$ , respectively.

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